Type MS · Getting terms out of a formula for the sum

Answer 1 of 6 · Question 6 part a · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1	
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1	
	$2\times17+d=38 \Rightarrow d=4$	A1	(4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1	(2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$		
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1	
	$2p^2-11p+12=0$	A1	
	$(2p-3)(p-4)=0 \Rightarrow p=4 \left(p\neq\frac{3}{2};\ \text{may not be seen}\right)$	M1A1	(4)
			[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $\left[a = S_1 = 1(15+2\times1) = 17\right]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}\left(2a+\left[n-1\right]d\right) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = \ldots$ and $30 = 2a-4 \Rightarrow a = \ldots$
		For the correct value of $d = 4$

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	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$\left(\quad 9\right)^2 \qquad 81 \quad \left(\quad 9\right)^2 \quad 25$	

Type MS · Getting terms out of a formula for the sum

Answer 2 of 6 · Question 6 part a · 2 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Mark
6(a)	$S_{20} = 20(3+2\times 20) = 860$	M1A1 [2]
(b)	$S_1 = 3\times 1+2\times 1^2 = 5$ $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$ $d = 9-5 = 4$ $U_n = '5'+(n-1)'4'$ $S_n = \sum_{r=1}^{n}(4r+1) \Rightarrow A = 4, \quad B = 1$	B1 M1 M1 A1 M1 A1 [6]
(c)	$T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right] \Rightarrow T_n = \frac{n}{2}(10+4n)$ or $T_n = n(5+2n)$ $n(5+2n) = 3n+2n^2+252, \quad 5n = 2n+252$ $5n = 3n+252 \Rightarrow n = 126$	M1 M1,dM1 ddM1A1 [5]
	Total	**13 marks**

Type MS · Getting terms out of a formula for the sum

Answer 2 of 6 · Question 6 part a · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
6(a)	M1	For substituting 20 into the given $S_{20} = 20(3+2\times 20) = 860$
	A1	For $S_{20} = 860$ Sight of 860 with no working scores M1A1
(b)	B1	For finding the first term $S_1 = 3\times 1+2\times 1^2 = 5$ or $a = 5$ Award this mark even if it is not clear that they understand that S_1 is the first term.
	M1	For a complete method to find the second term $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$
	M1	For finding the common difference, they must reach a value for this mark. $d = '9'-'5' = (4)$
	A1	For $d = 4$
	M1	For either $A = 4$ or $B = 1$ Accept embedded values.
	A1	For both $A = 4$ and $B = 1$ Accept embedded values.
	For the final correct answer seen without any or minimal working, award full marks in part (b)	
(c)	M1	For an expression for T_n using the given values $T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right]$
	M1	For equating their expression for T_n in $T_n = n(3+2n)+252$ The correct expression for S_n must be used here. $\Rightarrow \frac{n}{2}(10+4n) = n(3+2n)+252 \Rightarrow (5n = 3n+252)$ **This is an A mark in Epen**
	dM1	For forming a linear equation. e.g. $5n = 3n+252$ o.e. This mark is dependent on the previous M mark.
	ddM1	For solving **their** linear equation. $5n = 3n+252 \Rightarrow n = (126)$ This mark is dependent on both previous M marks.
	A1	For $n = 126$

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Answer 2 of 6 · Question 6 part a · 2 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Mark

Type MS · Getting terms out of a formula for the sum

Answer 3 of 6 · Question 7 part a.i · 0 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

7(a)	(i) $a = 9$	B1
	(ii) $d = 4$	B1 (2)
(b)	(i) $a = 4$	B1
	(ii) $r = 3$	B1 (2)
(c)	$A_{14} = \frac{14}{2}(2\times 9+13\times 4)$ or $\frac{14}{2}(9+61)$, $= 490$	M1, A1
	$"490"-6 = \frac{4(3^n-1)}{3-1}$	M1
	$3^n = 243$ $n = 5$	ddM1A1 (5) [9]

(a)	
B1	Correct value, no working or explanation needed
B1	Correct value, no working or explanation needed
(b)	
B1	Correct value, no working or explanation needed
B1	Correct value, no working or explanation needed
(c)	
M1	Use either formula for the sum of an arithmetic series with their a and d (if needed) and obtain a value for the sum of the first 14 terms
A1	Correct value for the sum
M1	Subtract 6 from their sum (explicitly or implicitly) and equate to the sum of the first n terms of the geometric series obtained using their a and r
ddM1	Solve their equation by a correct method. No method need be shown but must reach n = ... Depends on both M marks above
A1	Correct value for n obtained
ALT	For the last 3 marks:
M1	Subtract 6 from their sum and generate at least the first 5 terms of the geometric series.
ddM1	Sum their terms until at least “484” is reached
A1	Correct answer (5) obtained from correct work.

Type MS · Getting terms out of a formula for the sum

Answer 3 of 6 · Question 7 part a.i · 0 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Getting terms out of a formula for the sum

Answer 4 of 6 · Question 7 part a.ii · 2 marks

2019 JUN · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Getting terms out of a formula for the sum

Answer 5 of 6 · Question 7 part b.i · 0 marks

2019 JUN · Paper 02 · find part b.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Getting terms out of a formula for the sum

Answer 6 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 02 · find part b.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.